

# Objectives & Success Criteria

## Business Objectives

The primary objective of the proposed solution is to improve the efficiency, consistency, and scalability of medical imaging workflows by augmenting clinicians with AI-driven decision support capabilities.

The platform aims to:

- Reduce the time required to analyze complex medical imaging studies
  - Improve the quality and consistency of diagnostic assessments
  - Enhance clinicians' ability to understand complex anatomical structures through automated 3D reconstruction
  - Increase the throughput of imaging departments without proportionally increasing specialist staffing levels
  - Enable healthcare organizations to leverage AI while maintaining regulatory compliance and clinical governance
  - Establish a scalable foundation for future AI-driven diagnostic and predictive healthcare services
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## Operational Objectives

From an operational perspective, the solution should:

- Automate repetitive and time-consuming image preprocessing activities
- Automatically orchestrate specialized AI models based on study type and clinical context
- Integrate seamlessly with existing hospital systems (PACS, RIS, HIS)
- Minimize disruption to existing clinical workflows
- Provide clinicians with actionable outputs rather than raw AI predictions
- Ensure traceability of all AI-generated recommendations and transformations

## Technical Objectives

The platform should be designed to:

- Support hybrid deployment models (hospital edge infrastructure + cloud services)
- Process large DICOM datasets efficiently and reliably
- Orchestrate multiple AI services through a coordinated agent framework
- Support asynchronous and event-driven workflows
- Provide secure handling of sensitive medical data
- Enable future integration of additional AI models without architectural redesign
- Maintain high system availability and fault tolerance

## **AI-Specific Objectives**

The AI component of the solution should:

### **Image Understanding**

- Automatically identify relevant anatomical structures
- Generate accurate segmentations of target organs and tissues
- Detect candidate anomalies requiring clinician review

### **3D Reconstruction**

- Transform 2D imaging series into clinically useful 3D representations
- Improve visualization of complex anatomical relationships

### **Decision Support**

- Prioritize findings requiring urgent review
- Assist clinicians by highlighting areas of interest
- Provide confidence indicators and supporting evidence for AI-generated outputs

### **Continuous Improvement**

- Capture clinician feedback
- Enable supervised refinement of models over time
- Support future implementation of human-feedback-driven learning mechanisms

## **Success Criteria (KPIs)**

The following metrics will be used to evaluate the success of the Proof of Concept.

Clinical Efficiency KPIs

| Metric                                          | Baseline         | Target           |
|-------------------------------------------------|------------------|------------------|
| Average imaging study review time               | Current workflow | 20–40% reduction |
| Time required for 3D reconstruction preparation | Manual process   | >70% reduction   |
| Imaging department throughput                   | Current workload | 15–30% increase  |

Diagnostic Quality KPIs

| Metric                                   | Target                                          |
|------------------------------------------|-------------------------------------------------|
| Segmentation accuracy (Dice coefficient) | > 0.90                                          |
| Anatomical reconstruction quality        | Clinically acceptable in >95% of reviewed cases |
| Reduction in inter-observer variability  | 10–20%                                          |
| Clinician acceptance rate of AI outputs  | >80%                                            |

*Note: Exact thresholds may vary depending on the targeted clinical specialty and available validation datasets.*

Operational KPIs

| Metric                               | Target                      |
|--------------------------------------|-----------------------------|
| Successful DICOM processing rate     | >99%                        |
| End-to-end workflow completion rate  | >95%                        |
| AI orchestration failure rate        | <1%                         |
| Average processing latency per study | Within defined clinical SLA |

User Adoption KPIs

| Metric                                           | Target                  |
|--------------------------------------------------|-------------------------|
| Clinician satisfaction score                     | >4/5                    |
| Percentage of studies using AI-assisted workflow | >70% during pilot phase |
| Reduction in manual preprocessing tasks          | >50%                    |

### Compliance & Governance KPIs

| Metric                                    | Target |
|-------------------------------------------|--------|
| Traceability of AI decisions              | 100%   |
| Auditability of workflow events           | 100%   |
| GDPR compliance violations                | 0      |
| Security incidents involving patient data | 0      |

### PoC Success Criteria

Since this document describes a Proof of Concept, success is not measured solely by model accuracy.

The PoC will be considered successful if it demonstrates:

1. The feasibility of orchestrating multiple AI services through an agent-based architecture.
2. Successful integration with representative DICOM datasets.
3. Generation of clinically meaningful 3D reconstruction outputs.
4. Effective support of clinician review processes through AI-assisted insights.
5. Compliance with core security, privacy, and governance principles expected in healthcare environments.
6. A clear path toward production-scale deployment in a hybrid healthcare infrastructure.